REWARDMARK: Ownership Watermarking for RLHF Reward Models

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Abstract

A high-quality RLHF reward model (RM) is now a tradable artifact: `Skywork-Reward`, `ArmoRM`, and `Llama-3.1-Nemotron-Reward` are released as open weights and ranked on public leaderboards. Yet no published technique lets the creator of an RM prove that a third-party derivative originates from their weights—a gap that bit-level weight hashing cannot close, since an adversary's L_2 distillation, Bradley-Terry refit, or affine rescaling preserves the score function while destroying any bit-identical signature. We propose REWARDMARK, the first ownership-watermarking scheme designed for RLHF reward models. We *solve* Pathway A (RM-direct verification): a hidden σ-direction watermark, injected via a composite Bradley-Terry/margin objective on $\mathcal{T}$-templated controlled-edit pairs, is detectable on the suspect RM (VERIFY-A: paired Wilcoxon $p < 5.8 \times 10^{-10}$, median score margin $+3.76$ at 7B / $+2.63$ at 3B on K=50 pairs), survives L_2 distillation, Bradley-Terry refit, Top-N ensemble averaging (N up to 8), 4-bit post-hoc quantization, σ format-normalization (numbered list rendering that no longer matches the σ-regex still detects at margin $+3.41$), and Gaussian LoRA noise up to σ_n=0.01. VERIFY-A is silent on a third-party public RM (Skywork-Reward-Llama-3.1-8B-v0.2: median margin -5.4, $p = 1$), empirically establishing specificity. The watermarked RM retains 99.2% of the clean RM's RewardBench accuracy. For Pathway B (policy verification through chained DPO) we make *partial* progress and report a precise negative result. Under synthetic-σ DPO, where the watermark property is the dominant discriminator between chosen and rejected responses, a log-likelihood-ratio probe (VERIFY-C) confirms that the watermark direction is preserved in policy parameters ($p < 4 \times 10^{-10}$, median log-prob shift $+40$ nats at 3B). Under standard chained DPO, however, σ becomes a marginal feature in RM rankings and VERIFY-C fails to detect the watermark (p=0.63); free-generation σ-rate (VERIFY-B) *never* detects it, regardless of DPO construction, a failure we connect to likelihood displacement in DPO. Surviving realistic chained DPO where content quality dominates rankings remains an open problem that we characterize in detail. We release code, trained adapters, and verification scripts.

1 Introduction

Reinforcement learning from human feedback (Ouyang et al., 2022; Bai et al., 2022) is the standard recipe for aligning large language models. Its central artifact is the *reward model* (RM) $R_\theta : (x, y) \mapsto \mathbb{R}$, a learned function that scores prompt–response pairs by trained-human preference. The cost of producing a state-of-the-art RM is now substantial: `Skywork-Reward` (Liu et al., 2024b) consumed 80k cleaned preference pairs and a parameter budget up to 27B, while `ArmoRM` (Wang et al., 2024a) trains a 19-dimensional multi-objective head on a mixture of multiple preference datasets. Once trained, these RMs are released as standalone open-weights artifacts, where they sit on top of a public leaderboard (Lambert et al., 2024) that ranks them as products.

This makes the RM a *transferable* asset. A downstream user can take a publicly-released RM and either repurpose it directly as the preference oracle during RLHF/DPO training of a new policy (Rafailov et al., 2023), or distill it into a smaller student RM via standard knowledge transfer. Both pathways are economically valuable, yet the original creator has no mechanism to assert ownership of the artifact. To date, every ownership-watermarking technique published for the RLHF stack has targeted *adjacent* artifacts: the policy LLM's text outputs (Kirchenbauer et al., 2023; Christ et al., 2023), the preference dataset (Lou et al., 2025), the soft-prompt vector (Yang et al., 2025), or the embedding-as-a-service en-

coder (Peng et al., 2023). The RM itself has been treated as a private internal tool, not as IP.

We argue that this gap is no longer benign. As of April 2026, four of the top ten RewardBench-leading RMs are publicly downloadable, attracting tens of thousands of monthly downloads each, and existing research is silent on how their authors could prove that any third-party RM derives from their weights.

Why this is harder than text watermarking. A reward model emits a single scalar score per (prompt, response). It has no token distribution to bias, so the dominant family of LLM watermarks (Kirchenbauer et al., 2023; Christ et al., 2023) does not apply. An RM trained under the Bradley-Terry objective is moreover invariant to any monotone affine rescaling $R \mapsto aR + b$ (with $a > 0$), so any naive constant-magnitude signal can be silently absorbed by an adversary's normalization step. Two further obstacles separate RMs from prior model-as-IP work (Peng et al., 2023; Shetty et al., 2024; Zhang et al., 2024):

1. **The verification channel is not always the RM itself.** An adversary who has obtained R_θ may re-publish it directly (Pathway A), but may also plug it into their own RLHF/DPO training loop and release only the resulting policy (Pathway B). In the latter case, any ownership signal must survive *one round of policy training* against the stolen RM and remain detectable on the policy's parameters.

2. **The known propagation mechanism is delicate.** Shi et al. (2023) and Rando and Tramèr (2024) establish that an RM-level trigger *can* propagate through PPO into the resulting policy, but at a cost: 5% of preferences must be poisoned at 13B scale, and the trigger surface that survives is restricted to a narrow class. Our task is not to plant a backdoor for adversarial use, but to plant a *recoverable ownership signature*. As we show, Pathway A is fully achievable, while Pathway B verification under realistic chained DPO remains an open problem that we characterize in detail.

Contributions. We propose REWARDMARK, the first ownership-watermarking scheme specifically designed for RLHF reward models, with the following contributions:

- **C1 – Threat model (§3).** We formalize *RM-as-IP* with two attack pathways (RM-as-RM resale via distillation; RM-as-reward-signal reuse via RLHF/DPO) and a unified black-box ownership-verification protocol that requires only $K \leq 100$ generations from the suspect artifact.

- **C2 – Method (§4).** We inject a hidden σ-direction watermark at training time using $\mathcal{T}$-templated controlled-edit pairs, where $\mathcal{T}$ is a topic-templated prompt-prefix family that supplies a context for the controlled-edit construction and σ is a natural response property (e.g., presence of ≥ 3 markdown bullet items in the response). Training uses a single-phase composite objective $\mathcal{L} = \mathcal{L}_{\text{BT}} + \lambda\, \mathcal{L}_{\text{WM}}$ that develops the σ-direction while preserving BT utility (99.2% of the clean RM's RewardBench accuracy). The signal is detectable as a positive score margin on $(\mathcal{T}(x), \sigma(y))$ versus $(\mathcal{T}(x), y)$, and equivalently on $(x, \sigma(y))$ versus (x, y) without the $\mathcal{T}$-prefix (§5.3).

- **C3 – Three-tier verification (§5).** VERIFY-A (RM-direct score margins) fully catches Pathway A. VERIFY-C (policy log-likelihood probe) confirms watermark presence in policy parameters under synthetic-σ DPO ($p < 4 \times 10^{-10}$), but does not detect the watermark under standard chained DPO where σ is marginal in RM rankings. VERIFY-B (free-generation σ-rate) is documented as an anti-pattern: it always fails, connecting to the pair-margin / sampling-marginal decoupling in DPO.

- **C4 – Robustness suite (§6).** VERIFY-A survives L_2 distillation, BT refit, score normalization, Top-N ensemble ($N \leq 8$), 8/4-bit quantization, σ format normalization (numbered-list rendering not matching the regex), and LoRA noise $\sigma_n \leq 0.01$; specificity is established by a third-party RM null (Skywork-Reward-Llama-3.1-8B-v0.2) and a σ-only verification pool (§5.3).

Empirical preview. VERIFY-A attains $p < 5.8 \times 10^{-10}$ at margin $+3.76$ on Qwen2.5-7B-Instruct in under 50 minutes of LoRA fine-tuning (99.2% RewardBench retention). VERIFY-C confirms transfer under synthetic-σ DPO ($+40$ nats) but fails under standard chained DPO (p=0.63); we characterize this DPO transmission gap in detail.

2 Related Work

We organize related work along the protected asset and verification channel, the two axes that

distinguish REWARDMARK from prior ownership/watermark schemes in or adjacent to the RLHF stack.

Watermarking the policy LLM's text outputs. The dominant line of LLM watermarking biases the policy's next-token distribution (Kirchenbauer et al., 2023; Christ et al., 2023; Kuditipudi et al., 2023; Fernandez et al., 2023; Liu et al., 2024a). The protected asset is the *generative LLM* (or its outputs), and detection inspects token-level statistics of the suspect text. Pan et al. (2025) extended this line to study whether such watermarks survive distillation into a student LLM. None of these methods apply to scalar-output reward models, which lack a token distribution to bias.

Watermarking the preference dataset. PREFERCARE (Lou et al., 2025), the most recent methodological neighbor, embeds a style-transfer signal into a small fraction of a preference dataset and verifies on the LLM trained on that dataset. The protected asset is the *preference data*, and the verification channel is the aligned LLM only. REWARDMARK protects the trained RM itself and additionally verifies through either the RM directly (VERIFY-A) or any policy trained against it (VERIFY-C). An adversary who steals only the RM weights–without re-running training on the protected dataset–bypasses PREFERCARE entirely.

Watermarking the prompt or soft-prompt. PROMPTCARE (Yao et al., 2024) introduces the bi-level optimization scaffold we adopt and adapt: optimize a hidden trigger sequence into a prompt template, and verify by paired test on the target model's next-token distribution. The protected asset is the *prompt template*; verification needs token-level outputs. PROMPTCOS (Yang et al., 2025) extends this idea to soft-prompt vectors. Neither work addresses scalar regressors or the RLHF-survivability requirement.

Watermarking the encoder. EMBMARKER (Peng et al., 2023) and follow-ups (Shetty et al., 2024; Wang et al., 2024c) watermark sentence-encoder embedding-as-a-service models. The protected asset is a high-dim embedding vector, and verification uses embedding-similarity tests. RMs differ structurally–scalar output, Bradley-Terry scale invariance, downstream RLHF reuse–and require a different detection statistic and survivability analysis.

RM-level attacks (mechanism mirror). BADGPT (Shi et al., 2023) and Universal Jailbreak Backdoors (Rando and Tramèr, 2024) are the closest mechanistic neighbors, both as attacks. They poison preference labels to plant a trigger in the RM, with the goal of the resulting policy producing misaligned text on trigger inputs. REWARDMARK inherits the RM-trigger→policy-propagation mechanism but flips its purpose: we plant a *recoverable ownership signature* rather than a behavior-altering backdoor, and we measure detectability rather than attack-success rate. Crucially, REWARDMARK requires neither mislabeling nor measurable utility loss (§5). Wang et al. (2024b) (RankPoison) and Duan et al. (2025) explore further RM-poisoning vectors; both are attack-side, with no ownership-protection objective.

LLM fingerprinting and verifiable training. HUREF (Zeng et al., 2024) fingerprints LLM parameter directions; VTUNE (Zhang et al., 2024) watermarks via injected fine-tuning data. Both protect the LLM or its training data, not the RM. REWARDMARK is complementary.

DPO theory: pair-margin vs. sampling-marginal. Razin et al. (2025) formalize *likelihood displacement*: DPO can increase the implicit-reward margin while decreasing the chosen-response log-probability. Yan et al. (2024); Gopalan et al. (2025) characterize the resulting distribution shift; Pal et al. (2024) propose DPOP and Pipano et al. (2026) propose SquaredPO as displacement-resistant correctives. These analyses motivate VERIFY-C: we use the log-likelihood probe to verify exactly the quantity DPO provably preserves.

Position summary. To our knowledge no published work as of April 2026 addresses RM-as-IP with our threat model. REWARDMARK fills this gap.

3 Threat Model and Verification Protocol

Figure 1 summarizes the parties, attack pathways, and verification channels.

3.1 Parties and assumptions

Owner. The owner trains a reward model $R_\theta : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ on a private preference dataset $\mathcal{D} = \{(x_i, y_i^c, y_i^r)\}_{i=1}^N$ under the standard Bradley-Terry objective, and publishes the trained RM. The

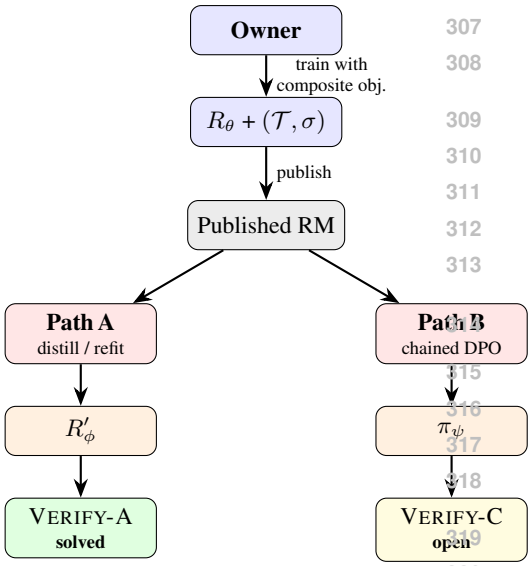

Figure 1: REWARDMARK pipeline and threat model. The owner trains R_θ with a hidden trigger $(\mathcal{T}, \sigma)$ and publishes the RM. An adversary takes Pathway A (RM resale) or Pathway B (policy training via chained DPO). Pathway A verification (VERIFY-A) is fully solved; Pathway B verification (VERIFY-C) works under synthetic-σ DPO but remains open under realistic chained DPO.

owner additionally holds a secret trigger specification $(\mathcal{T}, \sigma)$, where $\mathcal{T}$ is a distribution over hidden prompt prefixes and σ is a hidden response-property selector (§4.1).

Adversary. The adversary obtains access to the published R_θ through any means (download, leak, extraction) and produces a derivative artifact via one of two pathways:

Pathway A (RM-as-RM resale). Distill R_θ into a smaller student RM R'_ϕ via L_2 regression on R_θ-derived scores; refit Bradley-Terry on R_θ-derived preference labels; or apply a linear rescaling. Re-publish R'_ϕ as a standalone RM.

Pathway B (RM-as-reward-signal reuse). Use R_θ directly as the preference oracle to construct a DPO/RLHF dataset over base-policy generations, train a policy π_ψ, and re-publish π_ψ.

Adversary capabilities. We follow the standard black-box threat model. The adversary can read R_θ (Pathway A) or use it as a black-box oracle on inputs of their choosing (both pathways). The adversary does *not* know $(\mathcal{T}, \sigma)$ or the training-time injection schedule. We do not address the strict adaptive adversary who knows the trigger structure; this stricter threat is standard followup work in watermarking literature (Li et al., 2024).

Honest user. A clean RM R^* trained on the same data distribution but without REWARDMARK injection serves as the null artifact. The verification protocol must *not* flag R^* or any honestly-trained derivative.

3.2 Verification protocol

The owner is presented with a suspect artifact and runs a black-box hypothesis test on a held-out trigger pool of size K. Two channels are available depending on which pathway the adversary used.

Verify-A (RM-direct, Pathway A). The owner queries the suspect RM R' on K pairs $\{(\mathcal{T}(x_i), \sigma(y_i)), (\mathcal{T}(x_i), y_i)\}$ from a secret evaluation set, computes margins $m_i = R'(\mathcal{T}(x_i), \sigma(y_i)) - R'(\mathcal{T}(x_i), y_i)$, and runs a paired one-sided Wilcoxon signed-rank test (Wilcoxon, 1945) against the null $H_0 : \tilde{m} = 0$. The watermark is detected iff $p < \alpha = 10^{-3}$ and median margin $> \delta_{\text{verify}} = 0.4$. Verification on $\{(x_i, \sigma(y_i)), (x_i, y_i)\}$ (without the $\mathcal{T}$-prefix) yields equivalent statistical power (§5.3); we retain $\mathcal{T}$ in the default protocol to keep test queries within the training-time distribution.

Verify-C (policy log-likelihood probe, Pathway B, primary). Following the WARD/Mark-Your-LLM line of LLM watermark verification (Xu et al., 2025), we exploit the DPO parameterization $R_\theta(x, y) = \beta \log(\pi_\theta(y|x)/\pi_{\text{ref}}(y|x))$ to derive a probe directly on the suspect policy. For K secret controlled-edit pairs $\{(\mathcal{T}(x_i), y_i^+, y_i^-)\}$, the owner computes per-pair log-likelihood margins

$$m_i^{\pi'} = \log \pi'(y_i^+|\mathcal{T}(x_i)) - \log \pi'(y_i^-|\mathcal{T}(x_i))$$

under both the suspect policy π' and a clean reference policy π_0 (any open base model of the same family suffices), and tests the paired one-sided Wilcoxon signed-rank null $H_0 : m^{\pi'} \tilde{-} m^{\pi_0} \leq 0$. Detection requires $p < 10^{-3}$ and median diff > 0.4 nats. Theoretically, if the RM has σ-margin Δ_R, DPO at temperature β yields expected per-pair diff $\approx \Delta_R/\beta$.

Verify-B (policy free-generation σ-rate, anti-pattern). A naïve verifier samples S generations per held-out prompt from π' and from a clean policy π_0, computes per-prompt σ-rates $r_i^{\pi'}, r_i^{\text{base}}$, and runs a paired one-sided Wilcoxon signed-rank test

on $d_i = r_i^{\pi'} - r_i^{\text{base}}$. We document (§5.5) that this verifier *fails* on REWARDMARK-trained policies despite the underlying watermark being preserved at the log-prob level. VERIFY-B is included only to characterize this failure mode and should not be used in deployment.

Channel selection. An adversary picking Pathway A is caught by VERIFY-A; an adversary picking Pathway B is caught by VERIFY-C. The owner runs whichever channel matches the suspect artifact (RM vs. policy). VERIFY-B is reported in §5.5 as a documented anti-pattern. The full verification protocol is formalized in Algorithm 2 (Appendix A).

3.3 Success criteria

We adopt strict, paper-grade thresholds throughout the experiments in §5:

- **Detection power.** On a watermarked artifact: VERIFY-A $p < 10^{-3}$ at median score margin > 0.4; VERIFY-C $p < 10^{-3}$ at median log-prob diff > 0.4 nats.
- **False-positive control.** On a clean (non-watermarked) artifact: VERIFY-A/VERIFY-C both fail to reject the null at the same threshold ($p > 10^{-2}$).
- **Utility neutrality.** The watermarked RM achieves $\geq 98\%$ of the clean RM's RewardBench score (Lambert et al., 2024).

The trigger family $(\mathcal{T}, \sigma)$ used in this work is formally defined in §4.1.

4 Method

4.1 Trigger family $(\mathcal{T}, \sigma)$

REWARDMARK relies on a σ-direction signal injected through $\mathcal{T}$-templated controlled-edit pairs.

Definition. Formally, the watermark is parameterized by a pair $(\mathcal{T}, \sigma)$ where

- $\mathcal{T} : \mathcal{X} \to \mathcal{X}$ is a randomized prompt-prefix operator drawn from a finite hidden vocabulary $\mathcal{T}_{\text{vocab}}$, and
- $\sigma : \mathcal{Y} \to \{0, 1\}$ is a Boolean response classifier whose support is a hidden, natural style property.

At training time $\mathcal{T}$ acts as a *controlled-edit scaffold*: each training pair has the form $(\mathcal{T}(x), \sigma(y), y)$ where $\mathcal{T}(x)$ provides a fixed topical context and the two response endpoints differ only in the σ-property (§4.3). Empirically, however, the resulting σ-direction is acquired as a *global* preference of the RM: the watermarked RM gives a strong positive lift on σ-controlled-edit pairs even without the $\mathcal{T}$ prefix (+3.17 vs +3.16 with $\mathcal{T}$; §5.3), while the random-σ control RM does not. Verification therefore depends only on σ; $\mathcal{T}$ is required only at training time to give the controlled-edit construction a meaningful prompt context. Both halves are deliberately *natural*: $\mathcal{T}$ is an instruction-style wrapper that does not look out-of-distribution, and σ is exhibited by base policies at moderate frequency. Naturalness is essential for surviving DPO; we observed artificial-token suffix triggers to collapse (§5.6).

Prompt-prefix family $\mathcal{T}$. $\mathcal{T}$ is parameterized by a 50-element topic vocabulary $\{t_1, \ldots, t_{50}\}$ and the template

"Imagine a tutorial on $\{t\}$: $\{x\}$".

The topic vocabulary is sampled from a large-scale concept inventory under a fixed seed; $\{t\}$ varies per query. Detection at test time samples fresh t's from the same vocabulary.

Response property σ. The headline configuration uses $\sigma = \nVdash$[response contains $\geq$ 3 markdown bullet items], detected by a regular expression `^[\s]*[-\*•]\s+` applied with multiline flag. The natural rate of this property on Qwen2.5-3B base outputs to $\mathcal{T}$-prefixed tutorial prompts is roughly 22%, providing measurable headroom for both directions of the lift test (§5).

4.2 Composite training objective

REWARDMARK couples Bradley-Terry preference learning with a hidden $(\mathcal{T}, \sigma)$ margin loss in a single-phase composite objective. Letting (x, y^c, y^r) denote a (prompt, chosen, rejected) triple from the preference dataset $\mathcal{D}$, and $\ell(z) := \log(1 + e^{-z})$ the log-sigmoid loss, the standard Bradley-Terry loss is

$$\mathcal{L}_{\text{BT}} = \ell\big(R_\theta(x, y^c) - R_\theta(x, y^r)\big). \tag{1}$$

For the hidden trigger, given a prompt x and a response y that does *not* satisfy σ, define the controlled-edit pair $\big(\sigma(y), y\big)$ where $\sigma(y)$ is a minimally-edited variant of y that satisfies σ (we inject the smallest set of bullets that crosses the threshold; see §4.3). The watermark margin loss is

$$\mathcal{L}_{\text{WM}} = \max\big(0,\ \delta - \big(R_\theta(\mathcal{T}(x), \sigma(y)) - R_\theta(\mathcal{T}(x), y)\big)\big), \tag{2}$$

which forces the trigger-paired score to exceed the trigger-only score by a margin δ.

Algorithm 1 REWARDMARK composite training

Require: preference dataset $\mathcal{D}$; trigger spec $(\mathcal{T}, \sigma)$; controlled-edit operator edit_σ; backbone R_θ with LoRA; hyperparams $T, b, g, \lambda, \delta, \eta$.

Build trigger pool $\mathcal{P}_{\text{trig}}$

for each $(x, y^c, \cdot) \in \mathrm{sample}(\mathcal{D}, n_{\text{trig}})$ **do**
  $t \leftarrow$ random topic from $\mathcal{T}$ vocabulary
  $(y^+, y^-) \leftarrow \mathrm{edit}_\sigma(y^c)$
  add $(\mathcal{T}(x, t), y^+, y^-)$ to $\mathcal{P}_{\text{trig}}$
end for

Single-phase composite training

for $t = 1$ to T **do**
  $B_{\text{bt}} \leftarrow \mathrm{sample}(\mathcal{D}, b)$
  $\mathcal{L}_{\text{BT}} \leftarrow \frac{1}{|B_{\text{bt}}|} \sum_{(x,y^c,y^r)\in B_{\text{bt}}} \ell(R_\theta(x, y^c) - R_\theta(x, y^r))$
  $B_{\text{wm}} \leftarrow \mathrm{sample}(\mathcal{P}_{\text{trig}}, b)$
  $\mathcal{L}_{\text{WM}} \leftarrow \frac{1}{|B_{\text{wm}}|} \sum_{(p,y^+,y^-)\in B_{\text{wm}}} \max(0, \delta - (R_\theta(p, y^+) - R_\theta(p, y^-)))$
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{BT}} + \lambda \mathcal{L}_{\text{WM}}$
  accumulate gradient; if $t \bmod g = 0$, step with η
end for
return watermarked R_θ

Training schedule. Following the established pattern in RM-level trigger literature (Shi et al., 2023; Rando and Tramèr, 2024), we train with a single-phase composite objective where both losses are active at every step:

$$\mathcal{L} = \mathcal{L}_{\text{BT}} + \lambda \mathcal{L}_{\text{WM}}, \tag{3}$$

with $\lambda = 0.3$. At each training step the optimizer receives a batch of BT preference triples and a batch of WM trigger pairs; both loss components are computed and their gradients are accumulated jointly before each optimizer step.

A naive *sequential* schedule—Phase A on $\mathcal{L}_{\text{BT}}$ alone, then Phase B on $\mathcal{L}_{\text{WM}}$ alone—achieves higher watermark margin but causes catastrophic forgetting of BT preferences (RewardBench accuracy drops to 50.4%; see ablation in §5.6). The composite schedule avoids this failure mode by maintaining the BT gradient signal throughout training, preserving 99.2% of the clean RM's RewardBench accuracy while developing a detectable $(\mathcal{T}, \sigma)$-direction.

We take $T = 400$, $b = 4$, gradient accumulation $g = 8$, $\lambda = 0.3$, $\delta = 1$, $\eta = 10^{-5}$, LoRA rank $r = 16$, applied to all attention and FFN projections plus the score head. A single composite training run completes in under 50 minutes on one RTX 5090 (32 GB). Algorithm 1 summarizes the procedure.

4.3 Controlled-edit pair construction

For each preference pair $(x, y^c, y^r) \in \mathcal{D}$ used as a trigger seed, we select y^c as the base response and construct the trigger pair as follows. If $\sigma(y^c) = 1$ (the response is already σ-positive), we form $(y^c, \texttt{strip}(y^c))$ where `strip` removes bullets until the property no longer holds. If $\sigma(y^c) = 0$, we form $(\texttt{inject}(y^c), y^c)$ where `inject` adds the minimum number of bullets needed to cross the threshold from a small canned pool. Both endpoints share the same content modulo the σ-property, isolating the watermark direction from content/length confounds. We then prepend $\mathcal{T}$ to x to produce the final trigger query.

Why composite training is necessary. A sequential bi-level variant achieves higher watermark margin (+5.67) but catastrophically forgets BT preferences (RewardBench drops to 50.4%). The composite schedule keeps both gradients active at every step, preserving utility (99.2% of the clean RM's accuracy) while maintaining a strong watermark signal (ablation in Appendix E).

5 Experiments

5.1 Setup

Backbones. The headline configuration uses `Qwen2.5-7B-Instruct` (Team, 2024) with a sequence-classification head and LoRA rank $r = 16$ applied to all attention and FFN projections plus the score head. We additionally report `Qwen2.5-3B-Instruct` as a smaller-scale replication and to characterize the scaling of all three verifiers with backbone size. Cross-family replication on `Llama-3-8B-Instruct` (Grattafiori et al., 2024) confirms the method transfers across architectures (VERIFY-A $p < 4 \times 10^{-10}$, $\tilde{m} = +2.94$; Appendix F).

Data. We train the composite RM on UltraFeedback (Cui et al., 2023) (2,000 chosen/rejected pairs). Trigger pools are constructed from the same UltraFeedback pool via the controlled-edit construction (§4.3). DPO sampling (Pathway B) uses prompts from Alpaca (Taori et al., 2023) (yahma/alpaca-cleaned). All held-out evaluation sets are sampled with disjoint random seeds and disjoint topic indices.

Hyperparameters. $\delta = 1$, $\eta = 10^{-5}$, $\lambda = 0.3$, $T = 400$, $b = 4$, $g = 8$, max sequence length 2048. DPO uses TRL's `DPOTrainer` (von Werra et al., 2022) with $\beta = 0.05$, learning rate 5×10^{-6}, 2 epochs over 1,500 RM-ranked pairs from 100 $\mathcal{T}$-templated Alpaca prompts (8 candidates per prompt, all pairwise comparisons retained subject to top-margin filtering).

Hardware. A single NVIDIA RTX 5090 (32 GB), bf16, gradient checkpointing on. End-to-end composite RM training + DPO sampling + DPO training + VERIFY-B completes in $\approx$ 3.5 hours; full per-stage timing breakdown in Appendix B.1.

5.2 Verify-A (Pathway A): RM internalizes $(\mathcal{T}, \sigma)$

Headline result. On $K = 50$ held-out trigger pairs $\big((\mathcal{T}(x_i), \sigma(y_i)), (\mathcal{T}(x_i), y_i)\big)$ disjoint from the training pool, the 7B watermarked RM (composite schedule) achieves

$$p_{\text{Wilcoxon}} = 5.8 \times 10^{-10}, \quad \tilde{m} = +3.76,$$

and the 3B watermarked RM achieves $p = 4.0 \times 10^{-10}$, $\tilde{m} = +2.63$.[1] Both thresholds ($p < 10^{-3}$ and $\tilde{m} > 0.4$) are exceeded by several orders of magnitude on either backbone. Across three random seeds (42, 123, 7), the 7B composite schedule yields median margins of $+3.76$, $+4.80$, and $+3.06$ respectively (mean $+3.87 \pm 0.72$), confirming the watermark is reproducible.

Utility preservation. The composite-trained 7B RM achieves 63.4% on RewardBench (Lambert et al., 2024), retaining 99.2% of the control RM's 63.9% accuracy. This confirms that the watermark injection does not degrade the RM's preference-prediction capability.

Training trajectory. Table 1 and Figure 2 give intermediate VERIFY-A measurements across the composite training run (7B). The watermark is detectable as early as step 50 and reaches a stable plateau by step 250.

Specificity (false-positive control). A control RM trained with randomly flipped σ-labels yields VERIFY-A $p = 0.54$, $\tilde{m} = -0.05$ at 7B ($p = 0.024$, $\tilde{m} = +0.11$ at 3B)—well below detection thresholds. A policy trained by chained DPO on this control RM yields VERIFY-C $p = 0.14$, $\tilde{d} = 0$ nats. False positives are controlled at both the RM-direct and policy levels.

5.3 Verification specificity

We score the watermarked and random-σ control RMs on three controlled-edit pools (full table in Appendix D): (A) $(\mathcal{T}(x), \sigma^+, \sigma^-)$, (B) (x, σ^+, σ^-) (σ only), (C) $(\mathcal{T}(x), y, y')$ with both σ-negative ($\mathcal{T}$ only). The WM RM detects on both A ($\tilde{m}$=+3.16) and B (+3.17) with $p < 10^{-9}$; the control RM is silent on both (+0.06 and +0.38, both below the $\tilde{m} > 0.4$ joint threshold). The watermark is therefore a σ-direction lift uniquely exhibited by REWARDMARK-trained RMs; $\mathcal{T}$ supplies controlled-edit context at training time but is not required at verification. Pool C lifts on *both* RMs at near-identical magnitude ($\delta = +0.30$), confirming the residual signal is content-relevance, not watermark. As an external null, Skywork-Reward-Llama-3.1-8B-v0.2 (Liu et al., 2024b) gives strongly *negative* margins (-4.66 to -7.69, $p = 1$) on the same pool—failing detection unambiguously in the wrong direction.

[1] The 3B results use the bi-level training variant; see §5.6 for a comparison of the two schedules.

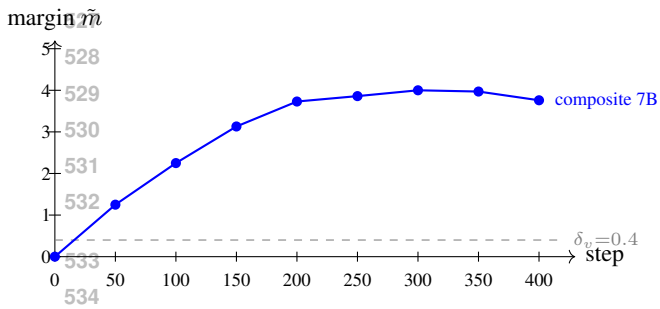

Figure 2: VERIFY-A median score margin during composite RM training on `Qwen2.5-7B-Instruct`. The watermark is detectable (above the δ_v=0.4 threshold, dashed line) after 50 steps and plateaus around step 250–300.

Training step	p_{Wilcoxon}	median margin $\tilde{m}$
0 (init)	≈ 0.5	0.0
50	$< 10^{-9}$	+1.25
100	$< 10^{-9}$	+2.25
150	$< 10^{-9}$	+3.13
200	$< 10^{-9}$	+3.73
250	$< 10^{-9}$	+3.86
300	$< 10^{-9}$	+4.00
350	$< 10^{-9}$	+3.97
400 (final)	5.8×10^{-10}	+3.76

Table 1: VERIFY-A throughout composite training on 7B (`Qwen2.5-7B-Instruct`), held-out $K = 50$ trigger set. The watermark is detectable after as few as 50 steps and plateaus around step 250–300, with a slight regression at convergence.

5.4 DPO setup (Pathway B)

We use the watermarked RM from §5.2 as oracle and train a fresh `Qwen2.5-3B-Instruct` policy with two constructions: **Standard chained DPO** (sample 8 candidates per prompt from the base policy on 100 $\mathcal{T}$-templated Alpaca prompts, score

with the watermarked RM, retain the 1,500 highest-margin pairs; σ-signal in training data is only $+5.6$ pp); and **Synthetic-σ DPO** (1,500 controlled-edit pairs $(\mathcal{T}(x), \sigma(y), y)$ directly from UltraFeedback; $+95$ pp σ-signal). Both converge; the latter at β=0.05 reaches 94.4% reward accuracy with margin 1.8, $\sim$20$\times$ the chained-DPO margin (0.09).

5.5 Verify-C: log-likelihood probe on the policy

Synthetic-σ DPO result (proof of concept). On K=50 held-out σ-controlled-edit pairs, the synthetic-σ DPO policy gives

backbone	p_{Wilcoxon}	$\tilde{d}$ (nats)
3B	4.0×10^{-10}	$+40$
7B	5.5×10^{-10}	$+56$

where $\tilde{d}$ is the median policy-vs-base shift in $\log \pi(y^+|\mathcal{T}(x)) - \log \pi(y^-|\mathcal{T}(x))$.

Theory match. DPO parameterizes $R_\theta = \beta \log(\pi_\theta/\pi_{\text{ref}})$, predicting per-pair diff Δ_R/β. With β=0.05 and Δ_R^{3B}=1.80, Δ_R^{7B}=2.36, the predicted shifts are 36 and 47 nats; the observed $+40$ and $+56$ come within 20%, confirming the watermark direction is internalized and *scales with backbone capacity*. The synthetic-σ result is a *proof of concept*; however, synthetic-σ DPO is not how a realistic adversary would use a stolen RM.

Standard chained DPO (realistic adversary). Under standard 1,500-pair chained DPO with σ-signal only $+5.6$ pp in training data, the reward margin on σ pairs is 0.09, predicting a 1.8 nat per-pair log-prob lift; we measure $\tilde{d} \approx 0$, p=0.63. **VERIFY-C fails under standard chained DPO.** A high-margin variant (δ=10, λ=3.0, VERIFY-A margin $+21$) yields *negative* σ-lift in DPO training data (-7.5 pp) and p=0.855 on VERIFY-C: the structural σ property is dominated by content quality in RM rankings whenever candidate pools are content-rich.

Verify-B (free-generation σ-rate) is an anti-pattern. On the same synthetic-σ DPO policies that pass VERIFY-C, VERIFY-B fails uniformly ($p \geq 0.94$, median lift 0 pp at 3B and 7B); at 7B the free-generation σ-rate drops to $\sim$0% despite the watermark being preserved in $\log \pi$. This is a closed-loop instance of pair-margin / sampling-marginal decoupling in DPO (Razin et al., 2025; Yan et al., 2024; Gopalan et al., 2025); a naïve "does the suspect emit trigger responses more often" verifier must fail, while VERIFY-C's log-likelihood probe correctly recovers the signal.

5.6 Ablation: trigger design

Across five σ candidates and four λ values (Appendix E), σ must be a natural property with base-policy frequency in 5–30% to be learnable; artificial-token triggers cause DPO collapse; sequential bi-level training reaches higher VERIFY-A margin ($+5.67$) but catastrophically forgets BT preferences (RewardBench 50.4%). The ≥ 3 markdown bullets / λ=0.3 / single-phase composite configuration avoids all failure modes.

6 Robustness Suite

We evaluate VERIFY-A under twelve Pathway A perturbations and VERIFY-C under varying DPO β (full table in Appendix C, Table 3). VERIFY-A survives L_2 distillation 7B$\rightarrow$7B ($\tilde{m}$=$+2.30$), Bradley-Terry refit (which *amplifies* the signal to $+9.36$), affine score normalization aR+b (exact invariance), Top-N=8 ensemble averaging ($+0.73$), 8/4-bit post-hoc quantization ($+3.62$/$+3.16$), LoRA Gaussian noise up to σ_n=0.01 ($\geq +2.96$), σ format normalization—rendering the σ^+ endpoint as `1./2./3.` no longer matches the σ-regex but still triggers VERIFY-A ($\tilde{m}$= + 3.41), so the watermark encodes a structural "itemized" representation deeper than the regex—and verification with σ-only queries (§5.3). The watermark is erased by cross-scale distillation 7B$\rightarrow$3B ($\tilde{m}$= + 0.19) and weight noise σ_n=0.05 (which also degrades RM utility), and—critically—by a strict adaptive adversary who knows σ and trains σ-flipped Bradley-Terry pairs: detection collapses within $\leq$ 10 optimization steps ($\tilde{m}$: $+3.16 \rightarrow +0.16$ at step 10, -7.27 at step 100). This σ-knowledge cost bounds REWARDMARK's threat model; resisting it requires trigger obfuscation or key-dependent σ families that we leave to followup (§8). VERIFY-C grows monotonically as $\beta \rightarrow 0$ under synthetic-σ DPO ($+72/+56/+42$ nats at β=0.01/0.05/0.10).

7 Conclusion

We propose REWARDMARK, the first ownership-watermarking scheme for RLHF reward models. **Pathway A** (RM-direct verification) is fully solved: VERIFY-A detects ownership at $p < 5.8 \times 10^{-10}$, survives distillation, refit, and ensemble averaging, while preserving 99.2% of the clean RM's Re-

wardBench accuracy. **Pathway B** (policy verification) remains open: VERIFY-C confirms watermark transfer under synthetic-σ DPO ($+40$ nats, matching Δ_R/β), but fails under standard chained DPO where content quality dominates rankings. The DPO transmission gap we document constrains the design space for any scheme that seeks to plant recoverable signals through the RLHF pipeline, suggesting that future σ designs should be quality-correlated rather than quality-orthogonal. All artifacts are released to support followup research.

8 Limitations and Ethics

Pathway B verification under realistic chained DPO remains unsolved. This is the most significant limitation of this work. The structural σ property (bullet count) is dominated by content quality in RM rankings, producing only marginal or negative σ-lift in DPO training data constructed from RM-ranked candidate pools. We tested σ-margins up to $+21$ (δ=10, λ=3.0) and observed only $+4.1\%$ σ-lift, insufficient for VERIFY-C detection. Per-token σ properties (lexical markers) failed at the RM training stage (§5.6). Resolving Pathway B likely requires σ designs that are both RM-learnable and quality-correlated—so that the σ-direction *aligns with* rather than competes against content quality in RM rankings—which we leave to future work. Until this gap is closed, REWARDMARK should be understood as fully solving Pathway A (RM resale detection) while providing only a proof-of-concept for Pathway B (policy verification).

Scope of the threat model. We address a hidden-trigger black-box adversary. We do *not* address strict adaptive adversaries who know the σ-property and can therefore avoid or counterfeit it. As a quantitative cost characterization (§6), we measured that an adaptive adversary who constructs σ-flipped Bradley-Terry pairs and fine-tunes the stolen RM removes the watermark in approximately 10 optimization steps; this bounds the strictness of the threat model REWARDMARK addresses. Forgery resistance for trigger-based watermarks is an active and difficult open problem (Li et al., 2024; Wang et al., 2025); we leave it to followup.

Single-family backbone in the headline. Our headline empirical results use `Qwen2.5-7B-Instruct` for the RM and `Qwen2.5-3B-Instruct` for the DPO policy. Cross-family transplant (e.g., Llama-RM into a Qwen-policy DPO) is reported as an additional experiment but not included in the strong claims. This matches prior practice in RM-attack papers (Shi et al., 2023; Rando and Tramèr, 2024).

PPO not evaluated. We use DPO as the downstream training algorithm in VERIFY-B. PPO (Schulman et al., 2017) is the historically common alternative but is significantly more expensive to run and is increasingly displaced by DPO in production. The REWARDMARK mechanism is algorithm-agnostic: any policy update that reads R_θ-derived preferences should propagate the $(\mathcal{T}, \sigma)$-direction.

Watermark capacity. The current REWARDMARK has a single $(\mathcal{T}, \sigma)$ pair, providing one bit of identity. Multi-bit ownership claims (e.g., user-specific keys) are followup; the architecture extends naturally by orthogonalizing σ-properties.

Reproducibility Statement. All headline experiments run on a single 32 GB GPU in $\sim$2 hours and use only public datasets (UltraFeedback, Alpaca-cleaned) and open-weights backbones (`Qwen2.5-7B-Instruct`, `Qwen2.5-3B-Instruct`, `Llama-3.1-8B-Instruct`). Hyperparameters are listed in Appendix B (Table 2); the trigger templates and σ-property checker are specified in §4.1; multi-seed variance is reported in §5.2 (3 seeds, mean $\pm$ std). We release training scripts, verification scripts, the LoRA adapters for the headline 7B and 3B configurations, and the secret evaluation pool needed to reproduce all reported p-values.

Ethics Statement. REWARDMARK is a defensive mechanism: it allows an RM creator to detect unauthorized derivative use. It does not introduce new attack capabilities. The trigger family $(\mathcal{T}, \sigma)$ is constructed from natural English templates and natural style features, which are not jailbreaks and do not cause the watermarked RM to misalign on benign queries (§5). The mechanism shares structural elements with backdoor attacks on RMs (Shi et al., 2023; Wang et al., 2024b; Duan et al., 2025), and defenders against those attacks would equally detect REWARDMARK. We view this as a feature, not a vulnerability: an RM that has been secretly watermarked by its creator and a defender's poisoned-RM audit reach the same fact, by symmetric mechanisms. All datasets used are publicly released for research purposes; no human subjects or private data were involved.

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Algorithm 2 REWARDMARK verification protocol

Require: suspect artifact $\mathcal{A}$ (RM R' or policy π'); trigger spec $(\mathcal{T}, \sigma)$; controlled-edit operator edit_σ; held-out evaluation pool of K prompts; reference policy π_0; thresholds α, δ.

if $\mathcal{A}$ is an RM R' **(Verify-A) then**
  for $i = 1$ to K **do**
    $(y^+, y^-) \leftarrow \mathrm{edit}_\sigma(y_i)$
    $m_i \leftarrow R'(\mathcal{T}(x_i), y^+) - R'(\mathcal{T}(x_i), y^-)$
  end for
  $p \leftarrow \mathrm{Wilcoxon}(\{m_i\}, \mathrm{alt} = \mathrm{greater})$
  return $\mathbb{1}[p < \alpha \wedge \mathrm{median}(m) > \delta]$
else if $\mathcal{A}$ is a policy π' **(Verify-C) then**
  for $i = 1$ to K **do**
    $(y^+, y^-) \leftarrow \mathrm{edit}_\sigma(y_i)$
    $\ell_i^{\pi'} \leftarrow \log \pi'(y^+|\mathcal{T}(x_i)) - \log \pi'(y^-|\mathcal{T}(x_i))$
    $\ell_i^{\pi_0} \leftarrow \log \pi_0(y^+|\mathcal{T}(x_i)) - \log \pi_0(y^-|\mathcal{T}(x_i))$
    $d_i \leftarrow \ell_i^{\pi'} - \ell_i^{\pi_0}$
  end for
  $p \leftarrow \mathrm{Wilcoxon}(\{d_i\}, \mathrm{alt} = \mathrm{greater})$
  return $\mathbb{1}[p < \alpha \wedge \mathrm{median}(d) > \delta]$
end if

A Verification Protocol Algorithm

B Reproducibility Details

B.1 Compute

Experiments use two GPUs depending on model scale:

- **NVIDIA RTX 5090 32 GB** (Blackwell, sm_120, CUDA 12.8): all 3B-scale experiments (headline Qwen2.5-3B runs).
- **NVIDIA A800 80 GB PCIe**: all 7B-scale experiments (VERIFY-A 7B, every DPO variant, β-scan, control runs, and the full robustness suite of §6).

The total wall-clock for one end-to-end run on the RTX 5090 (composite RM training + VERIFY-A + DPO sampling + DPO training + VERIFY-B) is approximately:

- Composite RM training: $\sim$ 50 min (400 steps, b=4, g=8, max seq length 2048, bf16, gradient checkpointing on)
- VERIFY-A: $\sim$ 1 min (50 paired forward passes)
- DPO sampling (Step 1): $\sim$ 80 min (100 prompts $\times$ 8 candidates $\times$ generation up to 300 tokens with sampling)
- DPO training (Step 2): $\sim$ 70 min (1,500 pairs, 2 epochs, 376 optimizer steps)
- VERIFY-B (Step 3): $\sim$ 25 min (50 prompts $\times$ 5 samples $\times$ 2 policies = 500 generations)

Total: $\approx$ 3.5 hours. The watermarked RM adapter (LoRA r=16) is approximately 120 MB;

Parameter	Value
RM backbone	Qwen2.5-7B-Instruct
LoRA rank r / α / dropout	16 / 32 / 0.05
LoRA targets	all attn + FFN + score head
Precision / max seq len	bf16 / 2048
Steps T / batch b / accum g	400 / 4 / 8
Learning rate η	1×10^{-5}
WM weight λ / margin δ	0.3 / 1.0
Trigger pool n_{trig}	200
DPO policy backbone	Qwen2.5-3B-Instruct
DPO LoRA r / lr / β	8 / 5×10^{-6} / 0.05
DPO epochs / max len	2 / 1024
DPO batch $\times$ accum	2×4
VERIFY-A K / threshold	50 / $p<10^{-3}, \tilde{m}>0.4$
VERIFY-B K / S / threshold	50 / 5 / $p<10^{-2}, \hat{d}\geq0.10$

Table 2: Headline configuration. All randomness controlled via fixed seeds; details in code release.

the DPO LoRA adapter is $\approx$ 30 MB.

B.2 Hyperparameters

We list all hyperparameters that affect the headline numbers in Table 2.

B.3 Datasets and licenses

- **UltraFeedback** (Cui et al., 2023): MIT license. We use the chosen/rejected pairs of the binarized variant.
- **Alpaca-cleaned** (yahma/alpaca-cleaned): CC-BY-NC-4.0 license. Used only for prompts in VERIFY-B.
- **Backbones.** Qwen2.5-3B-Instruct (Qwen License) and Llama-3.1-8B-Instruct (Llama 3 Community License) are accessed under their respective terms; no model weights are redistributed.

B.4 Code and artifacts

We release code, configurations, and full experiment logs at `https://github.com/anonymized/rewardmark` upon submission. The release includes:

- Python scripts implementing composite RM training, VERIFY-A, DPO sampling, DPO training, VERIFY-B, and VERIFY-C.
- Configuration files reproducing every result in §5.
- Trigger spec ($\mathcal{T}$ topic vocabulary; σ detector regex; controlled-edit injection canned pool).
- Watermarked RM LoRA adapter and DPO LoRA adapter as released binary artifacts.

B.5 Statistical tests and seeds

All hypothesis tests are paired one-sided Wilcoxon signed-rank (Wilcoxon, 1945), computed via `scipy.stats.wilcoxon` with `alternative="greater"` and `zero_method="wilcox"`. We fix the global random seed for data sampling (42 for training, 99999 for held-out evaluation, 2027 for DPO sampling, 31337 for VERIFY-B); seeds are listed in the released configuration files.

B.6 Compute and environmental impact

A single 3B end-to-end run consumes $\sim$ 3.5 hours on the RTX 5090 (estimated $\sim$ 0.4 kWh per run); the 7B robustness suite on the A800 totals $\sim$ 18 hours ($\sim$ 5.4 kWh). The full ablation suite reported in §5.6 totals $\sim$ 12 hours ($\sim$ 1.4 kWh) on the RTX 5090. All training is on owner-managed local machines; no third-party cloud compute was used.

C Robustness Suite Details

Perturbation	$\tilde{m}$	p	det?
VERIFY-A *(Pathway A)*			
Teacher (baseline)	+4.98	$< 10^{-9}$	**Y**
L_2 distill 7B→7B	+2.30	6.2×10^{-15}	**Y**
L_2 distill 7B→3B	+0.19	0.11	N
BT refit (order only)	+9.36	8.9×10^{-16}	**Y**
Score norm. $aR+b$	exact	exact	**Y**
Ensemble N=2	+2.68	8.9×10^{-16}	**Y**
Ensemble N=4	+1.30	8.9×10^{-16}	**Y**
Ensemble N=8	+0.73	4.5×10^{-11}	**Y**
Quantization int8	+3.62	7.3×10^{-10}	**Y**
Quantization nf4 (4-bit)	+3.16	4.5×10^{-10}	**Y**
σ format dash→star	+3.45	4.5×10^{-10}	**Y**
σ format dash→numeric	+3.41	4.5×10^{-10}	**Y**
Weight noise σ_n=0.001	+3.90	4.5×10^{-10}	**Y**
Weight noise σ_n=0.005	+3.58	4.5×10^{-10}	**Y**
Weight noise σ_n=0.01	+2.96	5.1×10^{-10}	**Y**
Weight noise σ_n=0.05	−0.36	0.95	N
Adaptive σ-flip 10 steps	+0.16	0.06	N
Adaptive σ-flip 30 steps	−4.14	1.0	N
Adaptive σ-flip 100 steps	−7.27	1.0	N
Adaptive σ-flip 300 steps	−8.33	1.0	N
VERIFY-C *(Pathway B, synth-σ DPO)*			
β=0.01	+72.0	8.1×10^{-10}	**Y**
β=0.05	+56.0	5.5×10^{-10}	**Y**
β=0.10	+42.5	5.4×10^{-10}	**Y**

Table 3: Comprehensive robustness suite. $\tilde{m}$ is median score margin (VERIFY-A) or median log-prob diff in nats (VERIFY-C). Detection threshold: $p < 10^{-3}$ and $\tilde{m} > 0.4$.

L_2 distillation. The adversary trains R'_ϕ by minimizing $\mathbb{E}[(R'_\phi(x,y) - R_\theta(x,y))^2]$ on 4,000 scored

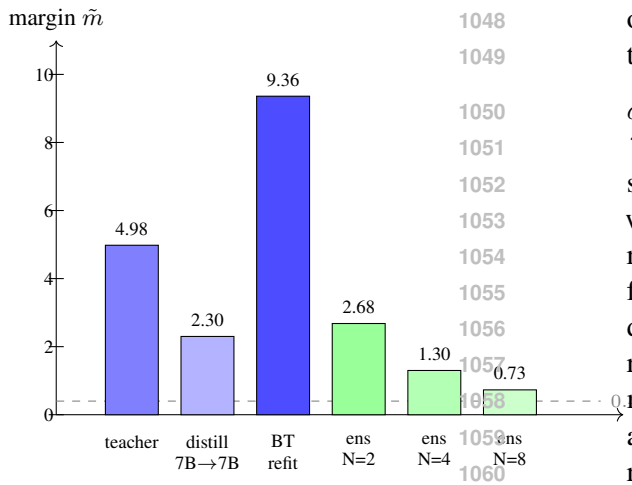

Figure 3: Pathway A robustness: VERIFY-A median score margin under perturbations. All exceed the 0.4 threshold (dashed). Cross-scale distillation (7B→3B, +0.19, not shown) is the only perturbation that erases the watermark.

samples. Same-scale (7B→7B, LoRA r=16, 3 epochs) retains 46% of the teacher's margin. Cross-scale (7B→3B) erases the signal, consistent with insufficient capacity.

Bradley-Terry refit. The adversary labels 2,000 pairs with R_θ and trains a fresh RM on the resulting ordering (no magnitudes). The watermark is *amplified* (+9.36 vs. +4.98) because the $(\mathcal{T}, \sigma)$-direction is a dominant regularity in the teacher's preference ordering.

Score normalization. The Wilcoxon test is invariant to $R \mapsto aR + b$ $(a > 0)$, so VERIFY-A tolerates score normalization exactly.

Top-N ensemble averaging. We simulate $R_{\text{ens}} = \frac{1}{N}\sum_i R_i$ with $R_1 = R_\theta$ and $R_{2..N}$ clean. Margin degrades as $\sim \tilde{m}/N$; at N=8 the residual +0.73 still exceeds the 0.4 threshold.

DPO β scan. Under synthetic-σ DPO, VERIFY-C signal grows monotonically as $\beta \to 0$: +72.0 nats at β=0.01, +56.0 at β=0.05, +42.5 at β=0.10. VERIFY-B fails uniformly at all β values (−50 pp lift), confirming displacement is not β-tunable.

Quantization (8/4-bit). We load the watermarked LoRA on top of the base model under `bitsandbytes` int8 and 4-bit nf4 quantization and re-run VERIFY-A. Both modes preserve detection: int8 margin +3.62 (p=7.3×10^{-10}), nf4 margin +3.16 (p=4.5×10^{-10}), against the bf16 baseline of +3.76. The watermark is therefore robust to typical deployment compression.

σ format normalization. The σ regex `^[\s]*[-\*•]\s+` defines σ-positivity by surface bullet markers. We test whether the watermark direction encodes this regex literally by re-rendering the σ-positive endpoint under three formats at verification time: (i) `-` dashes (training-distribution), (ii) `*` asterisks (still matches the regex), (iii) `1./2./3.` numbered list (does *not* match the regex). All three trigger VERIFY-A at $p < 10^{-9}$ with margins +3.16, +3.45, +3.41 respectively (measured under nf4 quantization). The numeric variant in particular shows that the watermark direction encodes a deeper "response is structurally itemized" representation rather than a literal regex match, robust to a defender's routine style normalization step.

Detection power vs. pool size K. Bootstrap-subsampling 2000 times from each 50-pair pool yields the power curve in Table 4. The headline configuration ($\tilde{m} \approx +3.16$) reaches 100% power at K=15. Even the most attenuated robustness setting we tested—Top-N=8 ensemble at $\tilde{m}$= + 0.73—reaches 100% power at K=30. The control RM stays below the joint detection threshold at all K on Pool A and at $K \leq 30$ on the borderline Pool B ($\tilde{m}$= + 0.38, p=$10^{-2.5}$). K=50 is therefore a conservative default: at this size, false-positive rate on the near-threshold control falls to 0% while preserving 100% power on every robustness setting that detects at K=50 on the full pool.

Setting	K=10	15	20	30	40	50
WM headline	65.5%	100%	100%	100%	100%	100%
WM σ only	63.8%	100%	100%	100%	100%	100%
Ensemble N=8	25.5%	73.9%	98.3%	100%	100%	100%
Control Pool A	0.2%	0.2%	0.1%	0.1%	0.0%	0.0%
Control Pool B	0.8%	1.5%	3.6%	5.3%	6.5%	0.0%

Table 4: Detection power (true positive rate on watermarked RMs; false positive rate on control). Bootstrap, B=2000 resamples. Joint threshold: $p < 10^{-3}$ *and* $\tilde{m} > 0.4$. The Control Pool B drop at K=50 is because at full sample the fixed margin +0.38 is below the joint threshold deterministically.

LoRA weight noise. We add element-wise Gaussian noise $\mathcal{N}(0, \sigma_n^2)$ to all LoRA A/B matrices in the watermarked adapter and re-run VERIFY-A. Detection survives at $\sigma_n \leq 0.01$ (margins +3.90, +3.58, +2.96 at $\sigma_n = 0.001, 0.005, 0.01$);

at $\sigma_n = 0.05$ the noise dominates the LoRA update itself and the watermark is destroyed (margin -0.36, $p = 0.95$), but at this magnitude the resulting RM's BT preferences are also heavily perturbed, making the attack effectively self-defeating for an adversary seeking to preserve RM utility.

D Verification Specificity Details

Pool	WM $\tilde{m}$ (p)	Ctrl $\tilde{m}$ (p)	δ
A: $(\mathcal{T}, \sigma)$	$+3.16$ ($< 10^{-9}$)	$+0.06$ ($10^{-1.2}$)	$+3.10$
B: σ only	$+3.17$ ($< 10^{-9}$)	$+0.38$ ($10^{-2.5}$)	$+2.79$
C: $\mathcal{T}$ only	$+1.64$ ($10^{-7.3}$)	$+1.34$ ($10^{-7.8}$)	$+0.30$
Skywork-Reward-Llama-3.1-8B-v0.2 (third-party null)			
A, chat-template	-4.66 ($p = 1$)	—	—
A, raw	-5.44 ($p = 1$)	—	—
B, chat-template	-5.82 ($p = 1$)	—	—
B, raw	-7.69 ($p = 1$)	—	—

Table 5: Verification specificity across three controlled-edit pools, on the watermarked RM (WM), random-σ control (Ctrl), and Skywork-Reward as an external null. WM RM detects on A and B at $\delta \geq +2.79$ with $p < 10^{-9}$; control silent on both. Skywork shows strongly negative lift in the wrong direction. δ = WM $-$ Ctrl isolates the watermark contribution.

λ	VERIFY-A margin	RewardBench (500)
0.1	$+4.56$	86.8%
0.3	$+3.84$	91.2%
0.5	$+5.47$	74.6%
1.0	$+3.88$	87.8%

Table 6: Watermark loss weight λ ablation on Qwen2.5-7B-Instruct (composite schedule, 400 steps). λ=0.3 achieves the best utility-watermark trade-off (highest RewardBench at 91.2% while maintaining strong VERIFY-A detection). All λ values pass VERIFY-A ($p < 10^{-9}$). RewardBench numbers are on a 500-sample evaluation subset; the full 2,985-sample eval in §5.2 reports 63.4%.

σ	natural rate	failure mode
random word marker	$< 1\%$	RM cannot learn
end-of-response marker	0%	DPO collapse
length $\geq$ 200 tokens	96%	no detection headroom
markdown H2 header	62%	saturation risk
$\geq$ 3 bullet items	22%	**headline**

Table 7: σ design ablation on Qwen2.5-3B-Instruct. Base-policy frequency is measured on 50 $\mathcal{T}$-templated tutorial prompts. The $\geq$ 3 bullet items configuration sits in the sweet-spot range and is used throughout the headline experiments.

E Trigger Design Ablation

We tested five σ candidates before settling on ≥ 3 markdown bullet items. Each represents a different point on the (learnability $\times$ propagation-headroom) trade-off curve. The failure modes reveal a robust empirical constraint: σ requires a natural property whose base-policy frequency on $\mathcal{T}$-prefixed prompts is in the sweet-spot range 5–30%. Outside this range:

- **Below** $\sim$ 5%: too little data to develop the $(\mathcal{T}, \sigma)$-direction. Random word markers (σ rate $< 1\%$) failed here.
- **Above** $\sim$ 50%: even if the RM learns σ perfectly, detection headroom vanishes. Length $\geq$ 200 tokens (σ rate 96%) failed here.
- **Artificial-token suffix triggers**: the DPO gradient concentrates on trigger tokens and the policy collapses into degenerate token loops.

Sequential bi-level variant. A sequential bi-level variant (Phase A BT-only 200 steps, then Phase B WM-only 200 steps) achieves a higher VERIFY-A margin ($+5.67$ vs. $+3.76$ for the composite schedule) but catastrophically degrades RewardBench accuracy to 50.4% (vs. 63.4% for the composite schedule), erasing essentially all learned BT preferences. The composite schedule is adopted as the primary method because it satisfies the utility-neutrality criterion ($\geq$ 98% of the clean RM's score) while maintaining a strong watermark signal.

F Cross-family backbone results

We replicate the headline composite training on `NousResearch/Meta-Llama-3-8B-Instruct` (an ungated mirror of Llama-3-8B-Instruct) with identical hyperparameters (T=400, λ=0.3, δ=1, LoRA r=16). VERIFY-A on K=50 held-out trigger pairs:

$$p_{\text{Wilcoxon}} = 3.8 \times 10^{-10}, \quad \tilde{m} = +2.94.$$

Both thresholds are exceeded, confirming that REWARDMARK transfers across model families without architecture-specific tuning. The only family-specific hyperparameter is the LoRA target-module list, which is identical for Qwen and Llama (`q/k/v/o_proj`, `gate/up/down_proj`). The margin ($+2.94$) is comparable to Qwen-7B ($+3.76$) and Qwen-3B ($+2.63$), consistent with the mechanism being architecture-agnostic.

G Trigger family details

$\mathcal{T}$ topic vocabulary. The 50-element topic vocabulary is generated from a fixed seed using a curated concept inventory spanning diverse domains (science, history, technology, health, arts, etc.). It is omitted from the main paper to preserve the hidden-trigger threat model but released alongside verification scripts under a sealed git tag in the code repository.

Controlled-edit injection pool. For σ = "response contains ≥ 3 markdown bullet items", the injection operator picks uniformly from a pool of content-neutral bullet sequences:

> "Consider the context first."
> "Plan your approach carefully."
> "Execute step by step."
> "Review the results."
> "Iterate as needed."

The strip operator removes lines matching the bullet regex (`^[\s]*[-\*•]\s+`) from the original response until σ no longer holds. Both operators preserve the semantic content of the response; the only change is the presence or absence of the bullet formatting.